AWS Step Functions — Important Interview Questions & Answers

1.What is AWS Step Functions?
AWS Step Functions is a serverless workflow service used to coordinate multiple AWS services and application steps.
It helps us build workflows where tasks run in a defined order with error handling and retries.
Simple ga gurthupettuko:
Step Functions ante multiple AWS services and tasks ni workflow laga connect chesi manage cheyyadaniki use chestham.
Eng: Step Functions is a serverless workflow service used to coordinate application tasks.

2.What is a State Machine?
A State Machine defines the complete workflow and the sequence of steps that should be executed.
It contains different states that represent individual steps in the workflow.
Simple ga gurthupettuko:
State Machine ante complete workflow and steps ni define chese configuration.
Eng: A State Machine defines the workflow and sequence of steps.

3.What is a State in Step Functions?
A State represents one step or activity in a Step Functions workflow.
Different states can perform tasks, make decisions, wait, run in parallel, or handle failures.
Simple ga gurthupettuko:
State ante workflow lo oka individual step.
Eng: A State represents an individual step in a workflow.

4.What is a Task State?
A Task State represents a unit of work performed by a workflow.
It can invoke services such as Lambda or other supported AWS service integrations.
Simple ga gurthupettuko:
Task State ante workflow lo actual work execute chese step.
Eng: A Task State represents a unit of work in a workflow.

5.What is a Choice State?
A Choice State is used to make decisions based on input data.
It allows the workflow to follow different paths depending on configured conditions.
Simple ga gurthupettuko:
Choice State ante condition batti workflow different path ki velladaniki use chestham.
Eng: A Choice State routes the workflow based on conditions.

6.What is a Parallel State?
A Parallel State allows multiple branches of a workflow to run at the same time.
It is useful when independent tasks can be executed simultaneously.
Simple ga gurthupettuko:
Parallel State ante independent tasks ni same time lo run cheyyadaniki use chestham.
Eng: A Parallel State runs multiple workflow branches simultaneously.

7.What is a Wait State?
A Wait State pauses the workflow for a specified amount of time or until a specific time.
It is useful when the workflow needs to wait before continuing to the next step.
Simple ga gurthupettuko:
Wait State ante workflow ni kontha time pause cheyyadaniki use chestham.
Eng: A Wait State pauses the workflow before continuing.

8.What is Retry in Step Functions?
Retry allows a failed task to be executed again based on configured retry conditions.
It is useful for handling temporary failures or service errors.
Simple ga gurthupettuko:
Retry ante task fail ayite configured conditions prakaram malli execute cheyyadam.
Eng: Retry allows a failed task to be executed again.

9.What is Catch in Step Functions?
Catch is used to handle errors when a task fails.
It allows the workflow to move to another state instead of completely failing.
Simple ga gurthupettuko:
Catch ante task fail ayinappudu error ni handle chesi alternate state ki move cheyyadam.
Eng: Catch handles task errors and moves the workflow to another state.

10.How does Step Functions work with Lambda?
Step Functions can invoke Lambda functions as part of a workflow.
The Lambda function performs a specific task and returns the result to the workflow for the next step.
Simple ga gurthupettuko:
Step Functions → Lambda → Result → Next Step.
Eng: Step Functions can invoke Lambda functions as workflow tasks.

11.What is the difference between Step Functions and Lambda?
Lambda is used to run application code without managing servers.
Step Functions is used to coordinate multiple tasks and services into a workflow.
Simple ga gurthupettuko:
Lambda code execute chestundi, Step Functions workflow ni manage chestundi.
Eng: Lambda runs code, while Step Functions orchestrates workflows.

12.Why do we use Step Functions?
We use Step Functions to coordinate multiple services, manage workflow execution, handle errors, and track workflow progress.
It is useful for applications that require multiple sequential or parallel steps.
Simple ga gurthupettuko:
Multiple steps and services ni proper workflow lo manage cheyyadaniki Step Functions use chestham.
Eng: Step Functions coordinates multiple tasks, handles errors, and manages workflow execution.

13.How would you handle a failed task in Step Functions?
I would configure Retry for temporary failures and Catch for handling errors that cannot be successfully retried.
Then I would route the workflow to an appropriate recovery or error-handling state.
Simple ga gurthupettuko:
Failure vaste Retry → still fail ayite Catch → error handling state.
Eng: I would use Retry for temporary failures and Catch for error handling.

14.What is the most important thing to remember about Step Functions?
Step Functions is mainly used to orchestrate and manage workflows involving multiple tasks and AWS services.
The important concepts are State Machines, States, Tasks, Choice, Parallel, Retry, and Catch.
Simple ga gurthupettuko:
Step Functions ante multiple tasks and AWS services ni workflow laga orchestrate cheyyadaniki use chese service.
Eng: Step Functions orchestrates workflows across multiple tasks and AWS services.